Deterministic native degree-83 training and holdout freeze

2026-10-12. Eight public synthetic fixtures were generated and independently replayed using standalone C++ field/curve arithmetic. This is a target/input freeze and correctness result; no benchmark was executed by these tools.

The question was to prepare three training and five holdout points for the exact retained degree-83, a=1, K600 comparison while avoiding target orbits already used by the prior observation or the retained base. The exact model is `icv1-f2m83-t6151469093347-9eba6cd5`, field `z^83+z^7+z^4+z^2+1`, curve $y^2+xy=x^3+x^2+1$, subgroup `r=8569786107849059`. The retained G was checked explicitly.

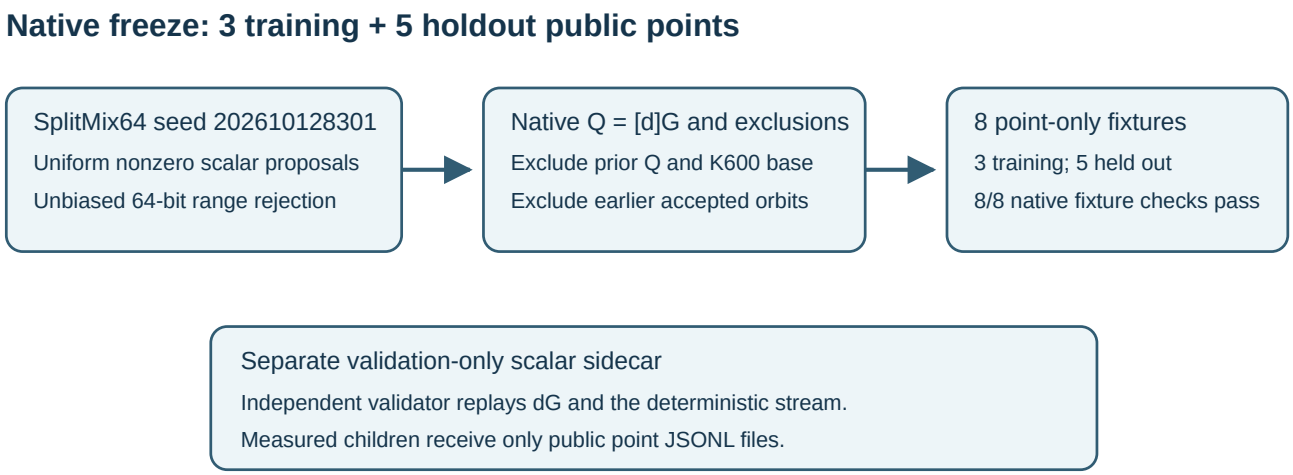


Figure 1: The scalar sidecar is for the independent validator only; measured children receive public point files.

Generation law and checks

The generator uses SplitMix64 seed `202610128301`. For bound `r-1`, it rejects 64-bit draws below `2^64 mod(r-1)` and maps accepted draws to `d=draw mod(r-1)+1`. This produces uniform nonzero subgroup scalar proposals without modulo bias. It computes `Q=[d]G`, rejects a candidate whose signed Frobenius orbit meets the prior fixed Q, K600 base or earlier accepted fixtures, and freezes the first three accepted points as training followed by five holdouts.

The canonical orbit key is the minimum polynomial-coordinate x among the 83 Frobenius shifts; negation leaves x unchanged. The native checker verified the field, subgroup and scalar eigenvalue and all retained base points/labels before using this exclusion key. The resulting accepted law is conditioned on the declared exclusions and avoids replacement of an orbit; this report does not call it an unconditioned IID sample.

Recorded counts:

Quantity	Recorded count
Training fixtures accepted	3
Holdout fixtures accepted	5
SplitMix64 draws	8
Uniform-range rejections	0
Excluded-orbit rejections	0
On-curve, subgroup and scalar point replays	8/8
Distinct signed orbits versus prior/base/one another	8/8

`check_optimization --fixtures-only` reproduced the full seed/rejection stream, all eight point/scalar identities, signed-orbit exclusions, individual point-file digests and exact five-row combined holdout file. The new receipt status is `PASS_FIXTURES_ONLY`; zero invalid rows. The independent executable uses the same standalone arithmetic helpers as

the generator and imports no research solver. It is independent of the subsequent Rust solver runs, not an independent implementation of every arithmetic primitive or an independent host.

Frozen files and remaining work

The eight individual JSONL files contain one `[x,y]` array each, with decimal coordinate strings and no scalar. The five-row holdout file uses the same point-only encoding. Scalars are retained separately in `validation_sidecar.json` and must not enter measured child inputs. The manifest records source/executable and retained-input hashes; external `sha256sum -c` confirmed every listed artifact. Generation and receipt creation fail closed on existing paths rather than overwriting evidence.

The generator receipt SHA-256 is:

`c93edc32a77411443229191e2db4cc3eeb7e99976b7f79a8a753ba103bce9909`

The independent fixture receipt SHA-256 is:

`fb263efde55e8177466ced20df6f09bd3b74c60b7c7505b62f91d6a650860315`

Native compile/replay commands and file layout are in `README.md`. The parent task will freeze the protocol before timings and later invoke the prepared output verifier after all timed children finish. No future scalar recovery, witness verification, timing ratio, statistical winner or full rank/factor-log transcript is claimed by this fixture-generation receipt. The existing historical model-alias correction remains explicit; no old receipt or target file was changed.